

Chapter 5

Direct Memory Access (DMA)



The direct memory access (DMA) module provides a hardware method of transferring data between peripherals and memory without intervention from the CPU; thereby, freeing up bandwidth for other system functions. Additionally, the DMA has the capability to orthogonally rearrange the data as the data is transferred as well as “ping-pong” data between buffers. These features are useful for structuring data into blocks for CPU processing.

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5.1 Introduction

The strength of a controller is not measured purely in processor speed, but in total system capabilities. As a part of the equation, any time the CPU bandwidth for a given function can be reduced, the greater the system capabilities. Many times applications spend a significant amount of the bandwidth moving data, whether moving data from off-chip memory to on-chip memory, from a peripheral such as an analog-to-digital converter (ADC) to RAM, or from one peripheral to another. Furthermore, many times this data comes in a format that is not conducive to the optimum processing powers of the CPU. The DMA module described in this chapter has the ability to free up CPU bandwidth and rearrange the data into a pattern for more streamlined processing.

The DMA module is an event-based machine, meaning the DMA module requires a peripheral or software trigger to start a DMA transfer. Although the DMA module can be made into a periodic time-driven machine by configuring a timer as the DMA trigger source, there is no mechanism within the module to start memory transfers periodically. The DMA module has six independent DMA channels that can be configured separately and each channel contains an independent PIE interrupt to let the CPU know when a DMA transfer has either started or completed. Five of the six channels are exactly the same, while Channel 1 has the ability to be configured at a higher priority than the others. At the heart of the DMA is a state machine and tightly coupled address control logic. This address control logic allows for rearrangement of the block of data during the transfer as well as the process of ping-ponging data between buffers. Each of these features is discussed in detail in this chapter.

5.1.1 Features

DMA features include:

- Six channels with independent PIE interrupts
- Each DMA channel can be triggered from multiple peripheral trigger sources independently
- Word Size: 16-bit or 32-bit (SPI limited to 16-bit)
- Throughput: 3 cycles/word without arbitration

5.1.2 Block Diagram

Figure 5-1 shows a device-level block diagram of the DMA.

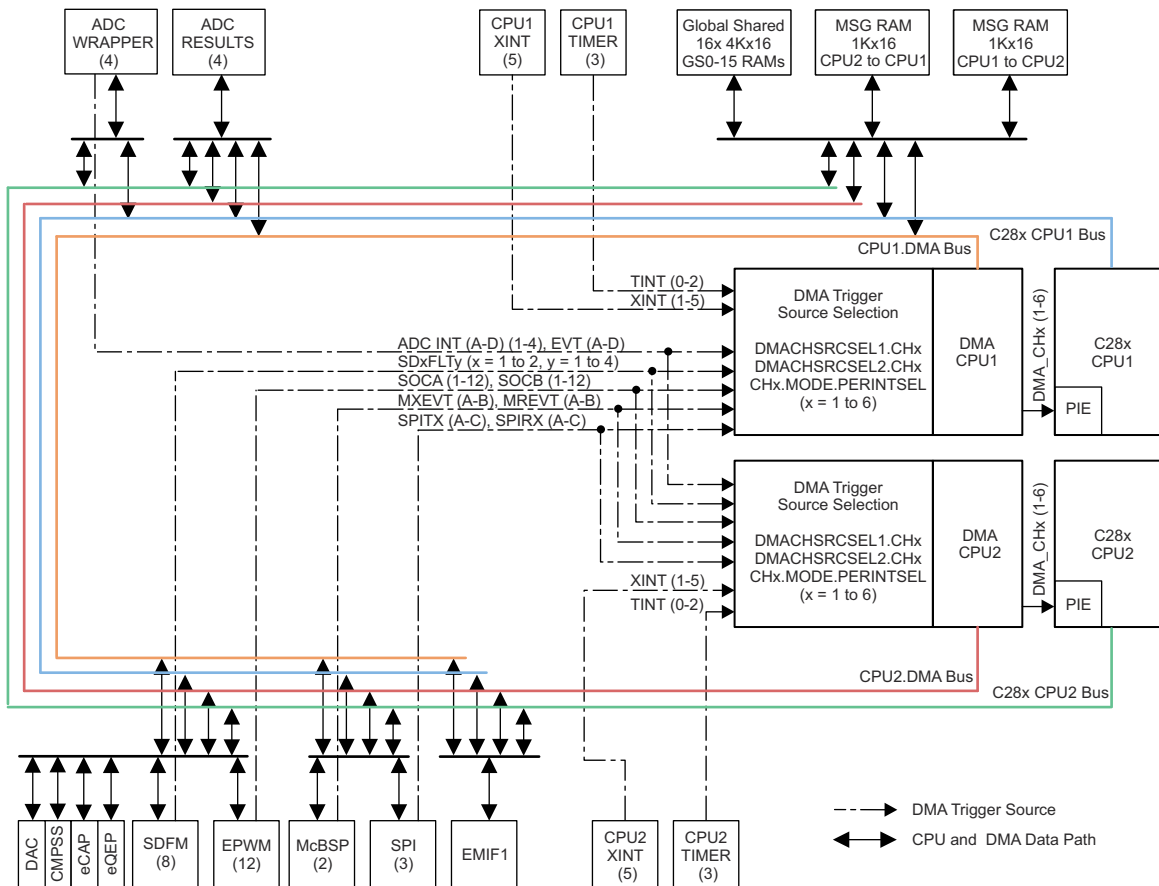


Figure 5-1. DMA Block Diagram

5.2 Architecture

5.2.1 Common Peripheral Architecture

There are two CPU subsystems; the CPU1 subsystem and CPU2 subsystem, with each containing a CLA and a DMA. The architecture allows several peripherals to be common between the two subsystems. Based on application need, these common peripherals can be attached to one of the two subsystems. Figure 5-2 shows how the CPUs and subsystems can be connected to the peripherals on peripheral frames 1 and 2. The clock, clock-enable, and reset muxing for the common peripherals are described in detail in other sections of this document.

Refer to Section 5.5 for more details on the arbitration scheme for all masters.

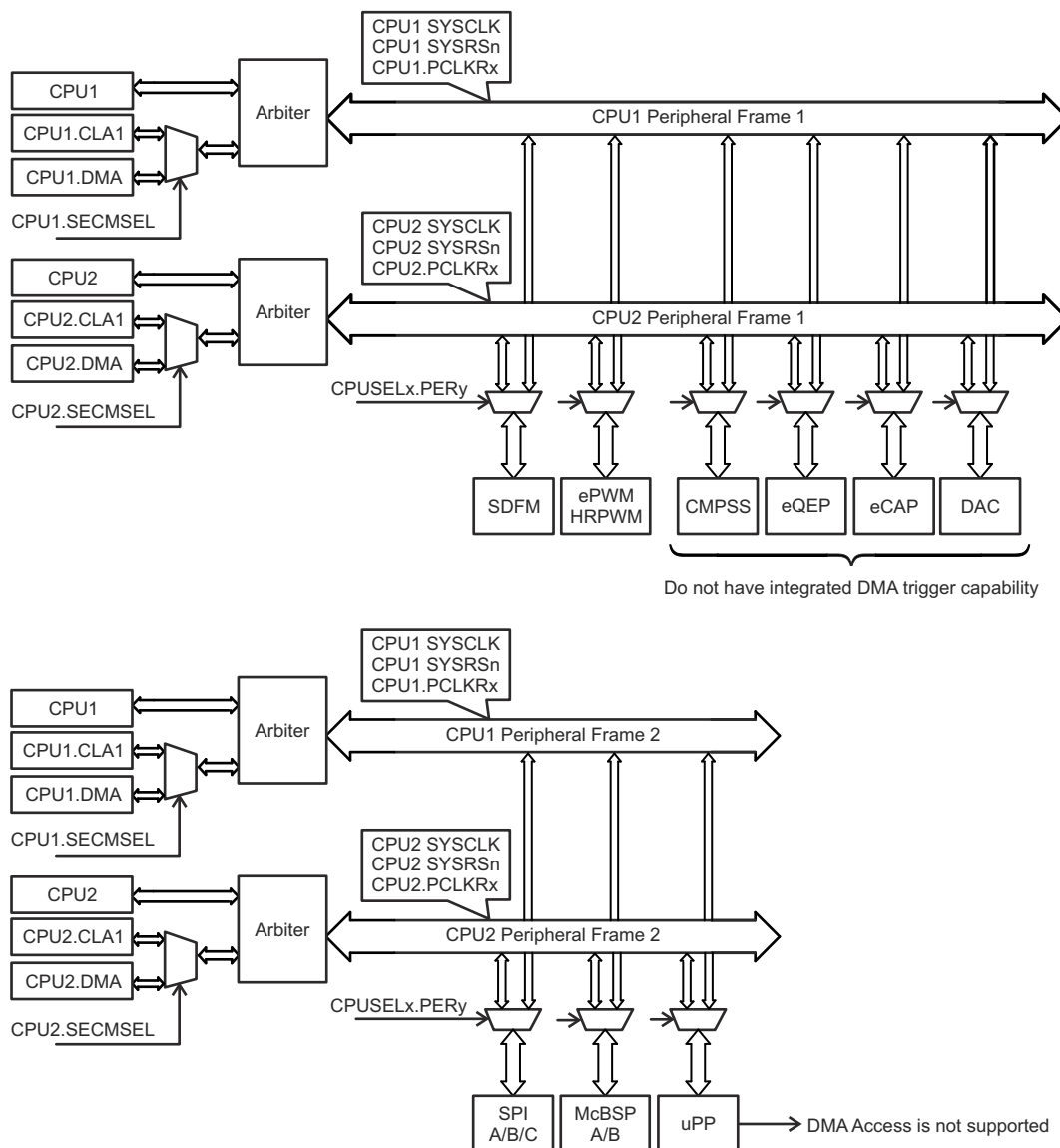


Figure 5-2. Common Peripheral Architecture

Note

If CPU and DMA make an access to the same peripheral frame in the same cycle, the DMA has priority and the CPU is stalled.

A CPUSEL bit associated with each peripheral defines whether the peripheral belongs to the CPU1 or CPU2 subsystem. If a peripheral belongs to a CPU subsystem, the peripheral can be accessed by the CPU and one of the secondary masters (DMA or CLA1). Refer to CPUSELx register definition for more details. The secondary master is statically selected using the SECMSEL register mapped to the respective CPU. Refer to CPUx.SECMSEL register definition for more details. If a secondary master is not selected, all writes from that master are ignored and all reads return 0x0 to any of the peripherals.

Similarly, if a peripheral does not belong to a CPU subsystem (as defined by the associated CPUSEL bit), all writes to that peripheral are ignored and all reads to that peripheral return 0x0 from any of the masters belonging to the unselected CPU subsystem. Note that since the arbiter has no knowledge regarding the ownership of individual peripherals (as can be seen from [Figure 5-2](#)), arbitration still happens even if the C28x or the selected secondary master tries to access a peripheral that does not belong to the CPU subsystem. See [Section 5.5](#) for more information.

5.2.2 Peripheral Interrupt Event Trigger Sources

Each DMA Channel can be configured to trigger by software and other peripheral triggers events. DMACHSRCSELx register can be used to configure DMA Trigger sources for each DMA channel. CHx.MODE.PERINTSEL register bit field can be set to channel number (CHx.MODE.PERINTSEL = x) as shown in [Figure 5-3](#). Included in these DMA Trigger sources are five external interrupt signals that can be connected to most of the general-purpose input/output (GPIO) pins on the device. This adds significant flexibility to the event trigger capabilities. Upon receipt of a peripheral interrupt event signal, the DMA automatically sends a clear signal to the interrupt source so that subsequent interrupt events occur.

Note

To use the system-level DMA Trigger source selection, the DMA internal trigger source selection configuration for each channel can be done using the DMACHSRCSELx register and the CHx.MODE.PERINTSEL register. See [Table 5-1](#) or the DMACHSRCSELx register definition for a complete list of DMA trigger sources.

Regardless of the value of the MODE.CHx[PERINTSEL] bit field, software can always force a trigger by using the CONTROL.CHx[PERINTFRC] bit. Likewise, software can always clear a pending DMA trigger using the CONTROL.CHx[PERINTCLR] bit.

Once a particular peripheral trigger event sets a channel's PERINTFLG bit, the bit remains pending until the priority logic of the state machine starts the burst transfer for that channel. Once the burst transfer starts, the flag is cleared. If a new peripheral trigger event is generated while a burst is in progress, the burst completes before responding to the new peripheral trigger event (after proper prioritization). If a third peripheral trigger event occurs before the pending event is serviced, an error flag is set in the CONTROL.CHx[OVRFLG] bit. If a peripheral trigger event occurs at the same time as the latched flag is being cleared, the trigger event has priority and the PERINTFLG remains set.

[Figure 5-4](#) shows a diagram of the trigger select circuit.

[Table 5-1](#) shows the peripheral trigger source options that are available for each channel.

CAUTION

See the Device Errata "ADC:DMA Read of Stale Result" Advisory regarding the potential for the DMA to read the ADC result registers before the result is ready

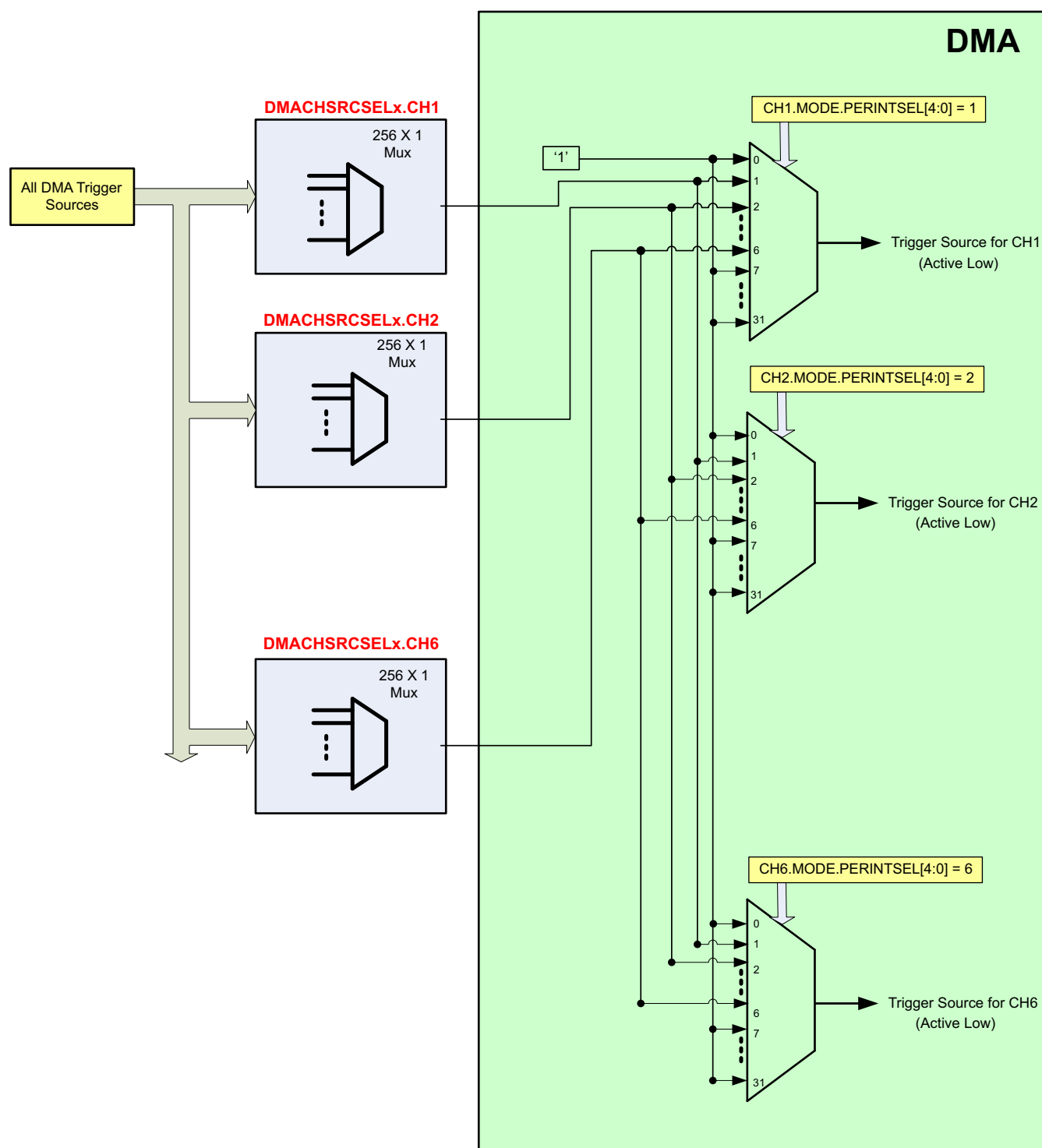
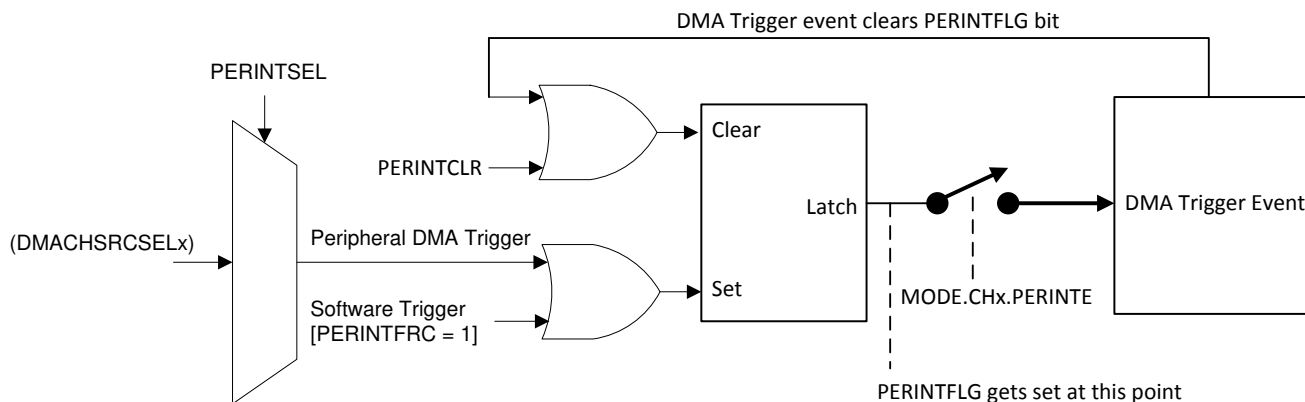


Figure 5-3. DMA Trigger Architecture



Note: See [Figure 5-3](#).

Figure 5-4. Peripheral Interrupt Trigger Input Diagram

Table 5-1. DMA Trigger Source Options

Select Index	Trigger Source
0	DMA_SOFTWARE_TRIGGER
1	ADCAINT1_DMA
2	ADCAINT2_DMA
3	ADCAINT3_DMA
4	ADCAINT4_DMA
5	ADCAEVT
6	ADCBINT1_DMA
7	ADCBINT2_DMA
8	ADCBINT3_DMA
9	ADCBINT4_DMA
10	ADCBEVT
11	ADCCINT1_DMA
12	ADCCINT2_DMA
13	ADCCINT3_DMA
14	ADCCINT4_DMA
15	ADCCEVT
16	ADCDINT1_DMA
17	ADCDINT2_DMA
18	ADCDINT3_DMA
19	ADCDINT4_DMA
20	ADCDEV
21-28	Reserved
29	XINT1
30	XINT2
31	XINT3
32	XINT4
33	XINT5
34-35	Reserved
36	EPWM1_SOCA
37	EPWM1_SOCB
38	EPWM2_SOCA

Table 5-1. DMA Trigger Source Options (continued)

Select Index	Trigger Source
39	EPWM2_SOCB
40	EPWM3_SOCA
41	EPWM3_SOCB
42	EPWM4_SOCA
43	EPWM4_SOCB
44	EPWM5_SOCA
45	EPWM5_SOCB
46	EPWM6_SOCA
47	EPWM6_SOCB
48	EPWM7_SOCA
49	EPWM7_SOCB
50	EPWM8_SOCA
51	EPWM8_SOCB
52	EPWM9_SOCA
53	EPWM9_SOCB
54	EPWM10_SOCA
55	EPWM10_SOCB
56	EPWM11_SOCA
57	EPWM11_SOCB
58	EPWM12_SOCA
59	EPWM12_SOCB
60-67	Reserved
68	CPU_TINT0
69	CPU_TINT1
70	CPU_TINT2
71	MCBSPA_XEVT
72	MCBSPA_REVT
73	MCBSPB_XEVT
74	MCBSPB_REVT
75-94	Reserved
95	SD1FLT1_DRINT
96	SD1FLT2_DRINT
97	SD1FLT3_DRINT
98	SD1FLT4_DRINT
99	SD2FLT1_DRINT
100	SD2FLT2_DRINT
101	SD2FLT3_DRINT
102	SD2FLT4_DRINT
103-108	Reserved
109	SPIA_TXDMA
110	SPIA_RXDMA
111	SPIB_TXDMA
112	SPIB_RXDMA
113	SPIC_TXDMA
114	SPIC_RXDMA

Table 5-1. DMA Trigger Source Options (continued)

Select Index	Trigger Source
115-126	Reserved
127	CLB1_INT
128	CLB2_INT
129	CLB3_INT
130	CLB4_INT
131-255	Reserved

5.2.3 DMA Bus

The DMA bus architecture consists of a 32-bit address bus, a 32-bit data read bus, and a 32-bit data write bus. Memories and register locations connected to the DMA bus by way of interfaces that sometimes share resources with the CPU memory or peripheral bus. Arbitration rules are defined in [Section 5.5](#).

5.3 Address Pointer and Transfer Control

The DMA state machine is, at the most basic level, two nested loops.

Burst (Inner) Loop:

The burst (inner) loop transfers a programmable number of words set by (BURST_SIZE + 1) register when a DMA channel trigger (Peripheral or Software trigger) is received. The BURST_SIZE register allows a maximum of 32 sixteen-bit words to be transferred in one burst. Each DMA channel supports both 16-bit or 32-bit word burst that can be controlled by MODE.DATASIZE bit field. Each DMA channel contains a shadowed address pointer for the source (SRC_ADDR_SHADOW) and the destination (DST_ADDR_SHADOW) address. At the beginning of each transfer, the shadowed version of each pointer is copied into the respective active (SRC_ADDR_ACTIVE or DST_ADDR_ACTIVE) register. During the burst loop, after each word is transferred, the signed value contained in the appropriate source or destination BURST_STEP register is added to the active register:

$$\text{SRC_ADDR_ACTIVE} = \text{SRC_ADDR_ACTIVE} + \text{SRC_BURST_STEP}$$

$$\text{DST_ADDR_ACTIVE} = \text{DST_ADDR_ACTIVE} + \text{DST_BURST_STEP}$$

The burst (inner) loop transfers a burst of data when a DMA Channel Trigger (Peripheral or Software trigger) is received.

Transfer (Outer) Loop:

The Transfer (outer) loop transfers a programmable number of bursts set by (TRANSFER_SIZE + 1) register for each channel. Since TRANSFER_SIZE is a 16-bit register, the total size of a transfer allowed is well beyond any practical requirement. During the transfer loop, after each burst is complete, there are two methods that can be used to modify the active address pointer:

Method 1 (Default): When address wrapping is disabled (SRC_WRAP_SIZE or DST_WRAP_SIZE is greater than TRANSFER_SIZE), active address pointer is updated as shown below

$$\text{SRC_ADDR_ACTIVE} = \text{SRC_ADDR_ACTIVE} + \text{SRC_TRANSFER_STEP}$$

$$\text{DST_ADDR_ACTIVE} = \text{DST_ADDR_ACTIVE} + \text{DST_TRANSFER_STEP}$$

Method 2: Address wrapping gets enabled when SRC_WRAP_SIZE or DST_WRAP_SIZE is less than TRANSFER_SIZE. This allows the channel to wrap multiple times within a single transfer. When the number of bursts is equal to (SRC/DST_WRAP_SIZE + 1) register, the state machine modifies the active address pointers as:

$$\text{SRC_BEG_ADDR_ACTIVE} = \text{SRC_BEG_ADDR_ACTIVE} + \text{SRC_WRAP_STEP}$$

$$\text{DST_BEG_ADDR_ACTIVE} = \text{DST_BEG_ADDR_ACTIVE} + \text{DST_WRAP_STEP}$$

$$\text{SRC_ADDR_ACTIVE} = \text{SRC_BEG_ADDR_ACTIVE}$$

$$\text{DST_ADDR_ACTIVE} = \text{DST_BEG_ADDR_ACTIVE}$$

At the end of DMA transfer, DMA can have transferred (BURST_SIZE + 1) x (TRANSFER_SIZE + 1) words.

OneShot Mode:

OneShot mode is disabled by default.

When OneShot mode is disabled (MODE.CHx[ONESHOT] = 0), DMA transfers one burst [(BURST_SIZE + 1) words] of data each time a DMA Channel Trigger is received. After the burst is completed, the state machine moves on to the next pending channel in the priority scheme, even if another trigger for the channel just completed is pending. This feature keeps any single channel from monopolizing the DMA bus.

When OneShot mode is enabled (MODE.CHx[ONESHOT] = 1), DMA transfers all the bursts [(BURST_SIZE + 1) x (TRANSFER_SIZE + 1) words] on a single DMA channel trigger. Be careful when using this mode, since this can create a condition where one trigger uses up the majority of the DMA bandwidth.

Continuous Mode:

Continuous mode is disabled by default.

When Continuous mode is disabled (MODE.CHx[CONTINUOUS] = 0), DMA state machine disables channel after all bursts in a transfer loop (TRANSFER_COUNT = 0) are complete. The channel must be re-enabled by setting the RUN bit in the CONTROL register before another transfer can be started on that channel.

When Continuous mode is enabled (MODE.CHx[CONTINUOUS] = 1), DMA state machine keep channel active even after all bursts in a transfer loop (TRANSFER_COUNT = 0) are complete.

Each DMA channel can trigger an EPIE interrupt for each DMA transfer either at start of DMA transfer or end of DMA transfer using MODE.CHx[CHINTMODE] bit.

Source/Destination Address Pointers (SRC/DST_ADDR) The value written into the shadow register is the start address of the first location where data is read or written to.

At the beginning of a transfer the shadow register (SRC/DST_ADDR_SHADOW) is copied into the active register (SRC/DST_ADDR_ACTIVE). The active register performs as the current address pointer.

Source/Destination Begin Address Pointers (SRC/DST_BEG_ADDR) This is the wrap pointer.

The value written into the shadow register (SRC/DST_BEG_ADDR_SHADOW) is loaded into the active register (SRC/DST_BEG_ADDR_ACTIVE) at the start of a transfer. On a wrap condition, the active register (SRC/DST_BEG_ADDR_ACTIVE) is incremented by the signed value in the appropriate SRC/DST_WRAP_STEP register prior to being loaded into the active register (SRC/DST_ADDR_ACTIVE).

For each channel, the transfer process can be controlled with the following size values:

**Source and
Destination
Burst Size
(BURST_SIZE)**

This specifies the number of words to be transferred in a burst.

This value is loaded into the BURST_COUNT register at the beginning of each burst. The BURST_COUNT decrements each word that is transferred and when the register reaches a zero value, the burst is complete, indicating that the next channel can be serviced. The behavior of the current channel is defined by the ONE_SHOT bit in the MODE register. The maximum size of the burst is dictated by the type of peripheral. For the ADC, the burst size can be all 16 registers (if all 16 registers are used). For RAM, the burst size can be up to the maximum allowed by the BURST_SIZE register, which is 32. See [Table 5-2](#) to understand how BURST_SIZE register affects the number of 16-bit words transferred with respect to DATASIZE.

Table 5-2. BURSTSIZE versus DATASIZE Behavior

BURSTSIZE	Number of 16-bit words transferred in	
	DataSize = 16-bit data	DataSize = 32-bit data
0	1	2
1	2	2
2	3	4
3	4	4
4	5	6
5	6	6
6	7	8
7	8	8
8	9	10
9	10	10
10	11	12
11	12	12
*	*	*
*	*	*
*	*	*
30	31	32
31	32	32

**Source and
Destination
Transfer Size
(TRANSFER_SIZE)**

This specifies the number of bursts to be transferred per CPU interrupt (if enabled).

Whether this interrupt is generated at the beginning or the end of the transfer is defined in the CHINTMODE bit in the MODE register. Whether the channel remains enabled or not after the transfer is completed is defined by the CONTINUOUS bit in the MODE register. The TRANSFER_SIZE register is loaded into the TRANSFER_COUNT register at the beginning of each transfer. The TRANSFER_COUNT register keeps track of how many bursts of data the channel has transferred and when the register reaches zero, the DMA transfer is complete.

Source/Destination Wrap Size (SRC/DST_WRAP_SIZE)	This specifies the number of bursts to be transferred before the current address pointer wraps around to the beginning. This feature is used to implement a circular addressing type function. This value is loaded into the appropriate SRC/DST_WRAP_COUNT register at the beginning of each transfer. The SRC/DST_WRAP_COUNT registers keep track of how many bursts of data the channel has transferred and when the registers reach zero, the wrap procedure is performed on the appropriate source or destination address pointer. A separate size and count register is allocated for source and destination pointers. To disable the wrap function, assign the value of these registers to be larger than the TRANSFER_SIZE.
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Note

The value written to the SIZE registers is one less than the intended size. So, to transfer three 16-bit words, the value 2 can be placed in the SIZE register.

Regardless of the state of the DATASIZE bit, the value specified in the SIZE registers are for 16-bit addresses. So, to transfer three 32-bit words, the value 5 can be placed in the SIZE register.

For each source/destination pointer, the address changes can be controlled with the following step values:

Source/Destination Burst Step (SRC/DST_BURST_STEP)	Within each burst transfer, the address source and destination step sizes are specified by these registers. This value is a signed 2s compliment number so that the address pointer can be incremented or decremented as required. If no increment is desired, such as when accessing the data receive or transmit registers in a communication peripheral, the value of these registers can be set to zero.
Source/Destination Transfer Step (SRC/DST_TRANSFER_STEP)	This specifies the address offset to start the next burst transfer after completing the current burst transfer. This is used in cases where registers or data memory locations are spaced at constant intervals. This value is a signed 2s compliment number so that the address pointer can be incremented or decremented as required.
Source/Destination Wrap Step (SRC/DST_WRAP_STEP)	When the wrap counter reaches zero, this value specifies the number of words to add/subtract from the SRC/DST_BEG_ADDR pointer and hence sets the new start address. This implements a circular type of addressing mode, useful in many applications. This value is a signed 2s compliment number so that the address pointer can be incremented or decremented as required.

Note

Regardless of the state of the DATASIZE bit, the value specified in the STEP registers are for 16-bit addresses. So, to increment one 32-bit address, a value of 2 can be placed in these registers.

Channel Interrupt Mode (CHINTMODE)	This mode bit selects whether the DMA interrupt from the respective channel is generated at the beginning of a new transfer or at the end of the transfer. If implementing a ping-pong buffer scheme with continuous mode of operation, then the interrupt can be generated at the beginning, just after the working registers are copied to the shadow set. If the DMA does not operate in continuous mode, then the interrupt is typically generated at the end when the transfer is complete.
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All of the previous features and modes are shown in [Figure 5-5](#). The following items are in reference to [Figure 5-5](#).

- The *HALT* points represent where the channel halts operation when interrupted by a high priority channel 1 trigger, or when the HALT command is set, or when an emulation halt is issued and the FREE bit is cleared to 0.
- The SRC/DST_ADDR_ACTIVE registers are not affected by SRC/DST_BEG_ADDR_ACTIVE at the start of a transfer. SRC/DST_BEG_ADDR_ACTIVE only affects the SRC/DST_ADDR_ACTIVE registers on a wrap. Following is what happens when a transfer first starts:
 - SRC/DST_BEG_ADDR_SHADOW remains unchanged.
 - SRC/DST_ADDR_SHADOW remains unchanged.
 - SRC/DST_BEG_ADDR_ACTIVE = SRC/DST_BEG_ADDR_SHADOW
 - SRC/DST_ADDR_ACTIVE = SRC/DST_ADDR_SHADOW
- The active registers get updated when a wrap occurs. The shadow registers remain unchanged. Specifically:
 - SRC/DST_BEG_ADDR_SHADOW remains unchanged.
 - SRC/DST_ADDR_SHADOW remains unchanged.
 - SRC/DST_BEG_ADDR_ACTIVE += SRC/DST_WRAP_STEP
 - SRC/DST_ADDR_ACTIVE = SRC/DST_BEG_ADDR_ACTIVE
- The best way to remember this is:
 - The shadow registers never change except by software.
 - The active registers never change except by hardware, and a shadow register is only copied into the active register, never an active register by another name.

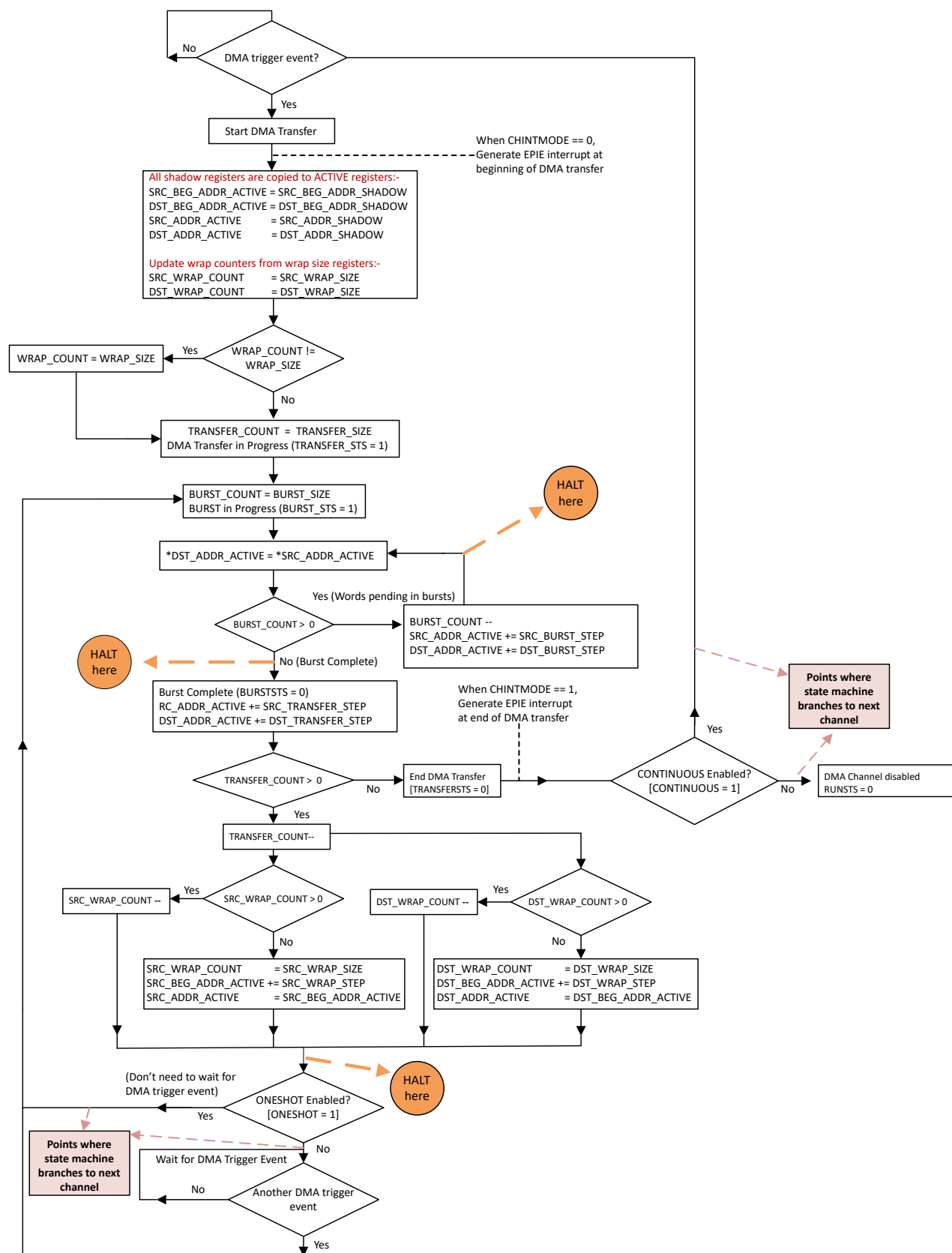


Figure 5-5. DMA State Diagram

5.4 Pipeline Timing and Throughput

In addition to the pipeline there are a few other behaviors of the DMA that affect the total throughput:

- A 1-cycle delay is added at the beginning of each burst
- A 1-cycle delay is added when returning from a CH1 high-priority interrupt
- Collisions with the CPU can add delay slots (see [Section 5.5](#))
- 32-bit transfers run at double the speed of a 16-bit transfer (takes the same amount of time to transfer a 32-bit word as to transfer a 16-bit word)

For example, to transfer 128 16-bit words from GS0 RAM to GS3 RAM, a channel can be configured to transfer 8 bursts of 16 words/burst. This gives:

$$8 \text{ bursts} * [(3 \text{ cycles/word} * 16 \text{ words/burst}) + 1] = 392 \text{ cycles}$$

If instead the channel were configured to transfer the same amount of data 32 bits at a time (the word size is configured to 32 bits), the transfer can take:

$$8 \text{ bursts} * [(3 \text{ cycles/word} * 8 \text{ words/burst}) + 1] = 200 \text{ cycles}$$

The DMA module consists of a 3-stage pipeline as shown in [Figure 5-6](#) and [Figure 5-7](#).

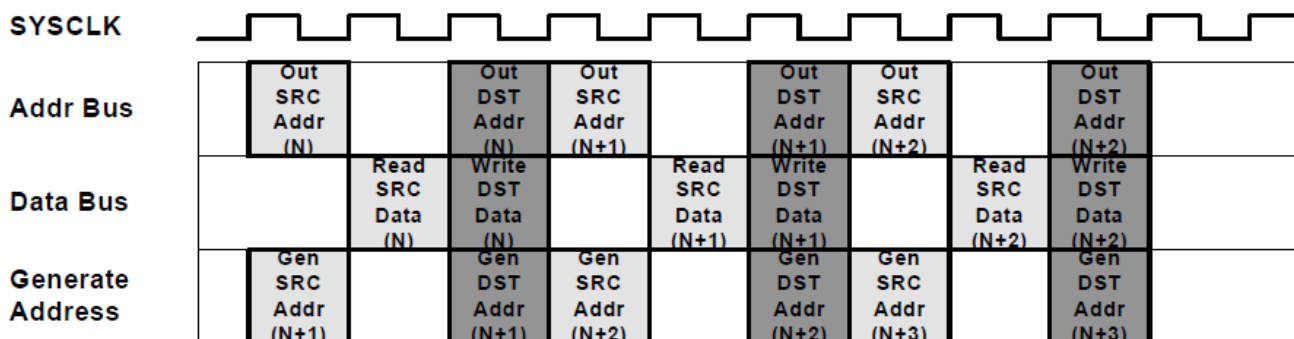


Figure 5-6. 3-Stage Pipeline DMA Transfer

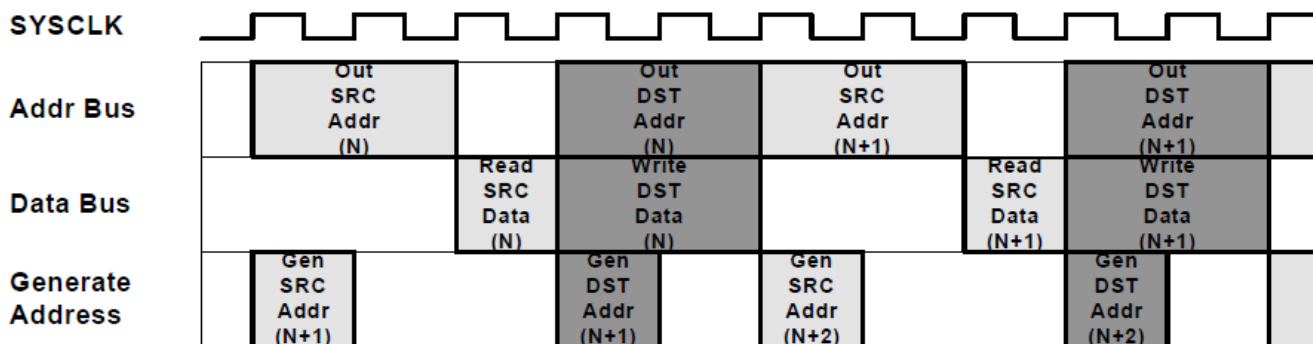


Figure 5-7. 3-stage Pipeline with One Read Stall

5.5 CPU and CLA Arbitration

Typically, DMA activity is independent of CPU and CLA activity. However, when the DMA and CPU (or CLA) try to access the same peripheral at the same time, an arbitration procedure is required to resolve the conflict. All instances of the same peripheral type conflict with each other. For instance, CAN-A and CAN-B conflict. Accesses to global shared RAM, across different instances, do not have this conflict. Different peripheral types can share a bus interface, which creates further opportunities for conflicts. These bus interfaces are:

- Peripheral frame 1: ePWM, eCAP, eQEP, CMPSS, DAC, SDFM
- Peripheral frame 2: PMBus, SPI

Conflict Example: The CLA is accessing DAC-A while the DMA is simultaneously accessing DAC-B.

Conflict Example: The CPU is accessing an SPI FIFO while the DMA is simultaneously accessing a PMBus register.

Non-conflict Example: The CPU is accessing a shared ePWM while the DMA is accessing an SPI.

Non-conflict Example: The CPU is accessing GS0 while the DMA is accessing GS1

The exception to all this is the ADC result registers, which are duplicated for each bus master. The CPU, DMA, and CLA can all simultaneously read these result registers with no stalls or arbitration needed for any master.

A DMA transfer consists of four phases: send source address, read source data, send destination address, and write destination data (see [Section 5.4](#)). Suppose CPU accesses a peripheral/memory causing conflict in middle of a DMA transfer, CPU is stalled till the current DMA access is complete and not until the completion of whole DMA transfer.

The following priority schemes are implemented for the various interfaces on the device:

- The fixed priority scheme for the peripheral frames is:
 - CLA/DMA Write
 - CLA/DMA Read
 - CPU Write
 - CPU Read
- The priority scheme for GSx RAM accesses is round-robin.
- The ADC results are duplicated for each CPU, CLA, and DMA so that no arbitration is needed when reading the result registers. This allows all masters to access the ADC result registers simultaneously without delay.

Note

If the CPU is performing a read-modify-write operation and the DMA performs a write to the same location, the DMA write can be lost if the operation occurs in between the CPU read and the CPU write. Avoid mixing CPU writes with DMA writes to the same locations.

Arbitration within DMA channels is based on a round-robin priority or Channel 1 high-priority scheme described in [Section 5.6](#).

5.6 Channel Priority

Two priority schemes exist when determining channel priority: Round-robin mode and Channel 1 high-priority mode.

5.6.1 Round-Robin Mode

In this mode, all channels have *equal* priority and each enabled channel is serviced in round-robin fashion as:

CH1 → CH2 → CH3 → CH4 → CH5 → CH6 → CH1 → CH2 → ...

In the case above, after each channel has transferred a burst of words, the next channel is serviced. The user can specify the size of the burst for each channel. Once CH6 (or the last enabled channel) has been serviced, and no other channels are pending, the round-robin state machine enters an idle state.

From the idle state, channel 1 (if enabled) is always serviced first. However, if the DMA is currently processing another channel *x*, all other pending channels between *x* and the end of the round are serviced before CH1. All the channels are of *equal* priority. For instance, take an example where CH1, CH4, and CH5 are enabled in round-robin mode and CH4 is currently being processed. Then CH1 and CH5 both receive an interrupt trigger from the respective peripherals before CH4 completes. CH1 and CH5 are now both pending. When CH4 completes the burst, CH5 is serviced next. Only after CH5 completes is CH1 serviced. Upon completion of CH1, if there are no more channels pending, the round-robin state machine enters an idle state.

A more complicated example is:

- Assume all channels are enabled, and the DMA is in an idle state,
- Initially a trigger occurs on CH1, CH3, and CH5 on the same cycle,
- When the CH1 burst transfer starts, requests from CH3 and CH5 are pending,
- Before completion of the CH1 burst, the DMA receives a request from CH2. Now the pending requests are from CH2, CH3, and CH5,
- After completing the CH1 burst, CH2 is serviced since this channel is next in the round-robin scheme after CH1.
- After the burst from CH2 is finished, the CH3 burst is serviced, followed by CH5 burst.
- Now while the CH5 burst is being serviced, the DMA receives a request from CH1, CH3, and CH6.
- The burst from CH6 starts after the completion of the CH5 burst, since this channel is the next channel after CH5 in the round-robin scheme.
- This is followed by the CH1 burst and then the CH3 burst
- After the CH3 burst finishes, assuming no more triggers have occurred, the round-robin state machine enters an idle state.

The round-robin state machine can be reset to the idle state using the DMACTRL[PRIORITYRESET] bit.

5.6.2 Channel 1 High-Priority Mode

In this mode, Channel 1 has high priority over all the other channels. Channels 2 to 6 have equal priority and each enabled channel is serviced in a round-robin fashion.

Higher priority: CH1

Lower priority: CH2 → CH3 → CH4 → CH5 → CH6 → CH2 → ...

Given an example where CH1, CH4, and CH5 are enabled in Channel 1 high-priority mode and CH4 is currently being processed. Then CH1 and CH5 both receive an interrupt trigger from the respective peripherals before CH4 completes. CH1 and CH5 are now both pending. When the current CH4 word transfer is completed, regardless of whether the DMA has completed the entire CH4 burst, CH4 execution is suspended and CH1 is serviced. After the CH1 burst completes, CH4 resumes execution.

Upon completion of CH4, CH5 is serviced. After CH5 completes, if there are no more channels pending, the round-robin state machine enters an idle state.

Typically Channel 1 is used in this mode for the ADC, since the data rate is so high. However, Channel 1 high-priority mode can be used in conjunction with any peripheral.

Note

High-priority mode and ONESHOT mode cannot be used at the same time on Channel 1. Other channels can use ONESHOT mode when Channel 1 is in high-priority mode.

5.7 Overrun Detection Feature

The DMA contains overrun detection logic. When a peripheral event trigger is received by the DMA, the PERINTFLG bit in the CONTROL register is set, pending the channel to the DMA state machine. When the burst for that channel is started, the PERINTFLG is cleared. If however, between the time that the PERINTFLG bit is set by an event trigger and cleared by the start of the burst, an additional event trigger arrives, the second trigger is lost. This condition sets the OVRFLG bit in the CONTROL register as in Figure 5-8. If the overrun interrupt is enabled, the channel interrupt is generated to the PIE module.

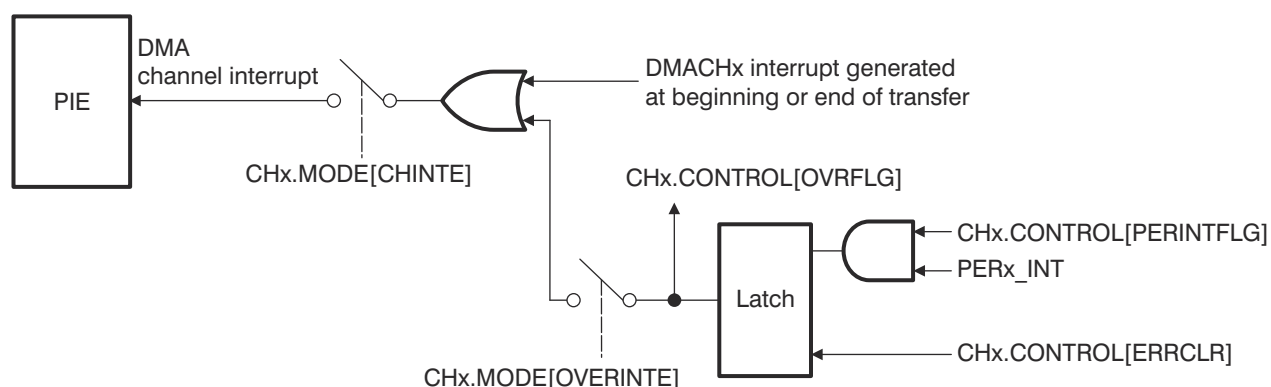


Figure 5-8. Overrun Detection Logic

5.8 Software

5.8.1 DMA Examples

NOTE: These examples are located in the [C2000Ware](#) installation at the following location:
C2000Ware_VERSION#/driverlib/DEVICE_GPN/examples/CORE_IF_MULTICORE/dma

Cloud access to these examples is available at the following link: dev.ti.com [C2000Ware Examples](#).

5.8.1.1 DMA GSRAM Transfer (dma_ex1_gsram_transfer)

FILE: dma_ex1_gsram_transfer.c

This example uses one DMA channel to transfer data from a buffer in RAMGS0 to a buffer in RAMGS1. The example sets the DMA channel PERINTFRC bit repeatedly until the transfer of 16 bursts (where each burst is 8 16-bit words) has been completed. When the whole transfer is complete, it will trigger the DMA interrupt.

: This example project has support for migration across our C2000 device families. If you are wanting to build this project from launchpad or controlCARD, please specify in the .syscfg file the board you're using. At any time you can select another device to migrate this example. *Watch Variables*

- *sData* - Data to send
- *rData* - Received data

5.8.1.2 DMA Transfer Shared Peripheral - C28X_DUAL

FILE: dma_ex1_shared_periph_cpu1.c

This example shows how to initiate a DMA transfer on CPU1 from a shared peripheral which is owned by CPU2. In this specific example, a timer ISR is used on CPU2 to initiate a SPI transfer which will trigger the CPU1 DMA. CPU1's DMA will then in turn update the ePWM1 CMPA value for the PWM which it owns. The PWM output can be observed on the GPIO pins. It is recommended to run the c28x1 core first, followed by the C28x2 core.

Watch Pins

- GPIO0 and GPIO1 - ePWM output can be viewed with oscilloscope

5.8.1.3 DMA Transfer for Shared Peripheral Example (CPU2) - C28X_DUAL

FILE: dma_ex1_shared_periph_cpu2.c

This example shows how to initiate a DMA transfer on CPU1 from a shared peripheral that is owned by CPU2. In this specific example, a timer ISR is used on CPU2 to initiate a SPI transfer that triggers the CPU1 DMA. CPU1 DMA then updates the ePWM1 CMPA value for the PWM that the DMA owns. The PWM output can be observed on the GPIO pins. It is recommended to run the c28x1 core first, followed by the C28x2 core.

5.8.1.4 DMA GSRAM Transfer (dma_ex2_gsram_transfer)

FILE: dma_ex2_gsram_transfer.c

This example uses one DMA channel to transfer data from a buffer in RAMGS0 to a buffer in RAMGS1. The example sets the DMA channel PERINTFRC bit repeatedly until the transfer of 16 bursts (where each burst is 8 16-bit words) has been completed. When the whole transfer is complete, it will trigger the DMA interrupt.

Watch Variables

- *sData* - Data to send
- *rData* - Received data

5.9 DMA Registers

This section describes the Direct Memory Access Registers.

5.9.1 DMA Base Addresses

Table 5-3. DMA Base Address Table

Device Registers	Register Name	Start Address	End Address
DmaRegs	DMA_REGS	0x0000_1000	0x0000_11FF

5.9.2 DMA_REGS Registers

Table 5-4 lists the memory-mapped registers for the DMA_REGS registers. All register offset addresses not listed in Table 5-4 should be considered as reserved locations and the register contents should not be modified.

Table 5-4. DMA_REGS Registers

Offset	Acronym	Register Name	Write Protection	Section
0h	DMACTRL	DMA Control Register	EALLOW	Go
1h	DEBUGCTRL	Debug Control Register	EALLOW	Go
4h	PRIORITYCTRL1	Priority Control 1 Register	EALLOW	Go
6h	PRIORITYSTAT	Priority Status Register	EALLOW	Go

Complex bit access types are encoded to fit into small table cells. Table 5-5 shows the codes that are used for access types in this section.

Table 5-5. DMA_REGS Access Type Codes

Access Type	Code	Description
Read Type		
R	R	Read
R-0	R -0	Read Returns 0s
Write Type		
W	W	Write
W1S	W 1S	Write 1 to set
Reset or Default Value		
-n		Value after reset or the default value

5.9.2.1 DMACTRL Register (Offset = 0h) [Reset = 0000h]

DMACTRL is shown in [Figure 5-9](#) and described in [Table 5-6](#).

Return to the [Summary Table](#).

DMA Control Register

Figure 5-9. DMACTRL Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED						PRIORITYRESET	HARDRESET
R-0h						R-0/W1S-0h	R-0/W1S-0h

Table 5-6. DMACTRL Register Field Descriptions

Bit	Field	Type	Reset	Description
15-2	RESERVED	R	0h	Reserved
1	PRIORITYRESET	R-0/W1S	0h	The priority reset bit resets the round-robin state machine when a 1 is written. Service starts from the first enabled channel. Writes of 0 are ignored and this bit always reads back a 0. When a 1 is written to this bit, any pending burst transfer completes before resetting the channel priority machine. If CH1 is configured as a high-priority channel, and this bit is written to while CH1 is servicing a burst, both the CH1 burst and the next pending low-priority burst are completed before the state machine is reset. If CH1 is high-priority, the state machine restarts from CH2 (or the next highest enabled channel). Reset type: SYSRSn
0	HARDRESET	R-0/W1S	0h	Writing a 1 to the hard reset bit resets the whole DMA and aborts any current access (similar to applying a device reset). Writes of 0 are ignored and this bit always reads back a 0. For a soft reset, a bit is provided for each channel to perform a gentler reset. Refer to the channel control registers. When writing to this bit, there is a one cycle delay before it takes effect. Hence, a one-cycle delay (such as a NOP instruction) is required in software before attempting to access any other DMA register. Reset type: SYSRSn

5.9.2.2 DEBUGCTRL Register (Offset = 1h) [Reset = 0000h]

DEBUGCTRL is shown in [Figure 5-10](#) and described in [Table 5-7](#).

Return to the [Summary Table](#).

Debug Control Register

Figure 5-10. DEBUGCTRL Register

15	14	13	12	11	10	9	8
FREE	RESERVED						
R/W-0h	R-0h						
7	6	5	4	3	2	1	0
RESERVED							
R-0h							

Table 5-7. DEBUGCTRL Register Field Descriptions

Bit	Field	Type	Reset	Description
15	FREE	R/W	0h	Emulation Control This bit specifies the action when an emulation halt event occurs. Reset type: SYSRSn 0h (R/W) = The DMA completes the current read-write operation, then halts. 1h (R/W) = The DMA continues running during an emulation halt.
14-0	RESERVED	R	0h	Reserved

5.9.2.3 PRIORITYCTRL1 Register (Offset = 4h) [Reset = 0000h]

PRIORITYCTRL1 is shown in [Figure 5-11](#) and described in [Table 5-8](#).

Return to the [Summary Table](#).

Priority Control 1 Register

Figure 5-11. PRIORITYCTRL1 Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED							CH1PRIORITY
R-0h							R/W-0h

Table 5-8. PRIORITYCTRL1 Register Field Descriptions

Bit	Field	Type	Reset	Description
15-1	RESERVED	R	0h	Reserved
0	CH1PRIORITY	R/W	0h	DMA Channel 1 Priority This bit selects whether CH1 has high priority or not. The priority can only be changed when all channels are disabled. A priority reset should be performed before restarting channels after changing priority Reset type: SYSRSn 0h (R/W) = CH1 has the same priority as the other channels 1h (R/W) = CH1 has a higher priority than the other channels

5.9.2.4 PRIORITYSTAT Register (Offset = 6h) [Reset = 0000h]

PRIORITYSTAT is shown in [Figure 5-12](#) and described in [Table 5-9](#).

Return to the [Summary Table](#).

Priority Status Register

Figure 5-12. PRIORITYSTAT Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED	ACTIVESTS_SHADOW			RESERVED	ACTIVESTS		
R-0h	R-0h			R-0h	R-0h		

Table 5-9. PRIORITYSTAT Register Field Descriptions

Bit	Field	Type	Reset	Description
15-7	RESERVED	R	0h	Reserved
6-4	ACTIVESTS_SHADOW	R	0h	Active Channel Status Shadow These bits are only meaningful when CH1 is in high-priority mode. When CH1 is serviced, the ACTIVESTS bits are copied to the shadow bits and indicate which channel was interrupted by CH1. When CH1 service is completed, the shadow bits are copied back to the ACTIVESTS bits. If this bit field is zero or the same as the ACTIVESTS bit field, then no channel is pending due to a CH1 interrupt. When CH1 is not a higher priority channel, these bits should be ignored. Reset type: SYSRSn 0h (R/W) = No channel is active 1h (R/W) = CH 1 2h (R/W) = CH 2 3h (R/W) = CH 3 4h (R/W) = CH 4 5h (R/W) = CH 5 6h (R/W) = CH 6 7h (R/W) = Reserved
3	RESERVED	R	0h	Reserved
2-0	ACTIVESTS	R	0h	Active Channel Status These bits indicate which channel (if any) is currently active or performing a transfer. Reset type: SYSRSn 0h (R/W) = No channel is active 1h (R/W) = CH 1 2h (R/W) = CH 2 3h (R/W) = CH 3 4h (R/W) = CH 4 5h (R/W) = CH 5 6h (R/W) = CH 6 7h (R/W) = Reserved

5.9.3 DMA_CH_REGS Registers

Table 5-10 lists the memory-mapped registers for the DMA_CH_REGS registers. All register offset addresses not listed in Table 5-10 should be considered as reserved locations and the register contents should not be modified.

Table 5-10. DMA_CH_REGS Registers

Offset	Acronym	Register Name	Write Protection	Section
0h	MODE	Mode Register	EALLOW	Go
1h	CONTROL	Control Register	EALLOW	Go
2h	BURST_SIZE	Burst Size Register	EALLOW	Go
3h	BURST_COUNT	Burst Count Register	EALLOW	Go
4h	SRC_BURST_STEP	Source Burst Step Register	EALLOW	Go
5h	DST_BURST_STEP	Destination Burst Step Register	EALLOW	Go
6h	TRANSFER_SIZE	Transfer Size Register	EALLOW	Go
7h	TRANSFER_COUNT	Transfer Count Register	EALLOW	Go
8h	SRC_TRANSFER_STEP	Source Transfer Step Register	EALLOW	Go
9h	DST_TRANSFER_STEP	Destination Transfer Step Register	EALLOW	Go
Ah	SRC_WRAP_SIZE	Source Wrap Size Register	EALLOW	Go
Bh	SRC_WRAP_COUNT	Source Wrap Count Register	EALLOW	Go
Ch	SRC_WRAP_STEP	Source Wrap Step Register	EALLOW	Go
Dh	DST_WRAP_SIZE	Destination Wrap Size Register	EALLOW	Go
Eh	DST_WRAP_COUNT	Destination Wrap Count Register	EALLOW	Go
Fh	DST_WRAP_STEP	Destination Wrap Step Register	EALLOW	Go
10h	SRC_BEG_ADDR_SHADOW	Source Begin Address Shadow Register	EALLOW	Go
12h	SRC_ADDR_SHADOW	Source Address Shadow Register	EALLOW	Go
14h	SRC_BEG_ADDR_ACTIVE	Source Begin Address Active Register	EALLOW	Go
16h	SRC_ADDR_ACTIVE	Source Address Active Register	EALLOW	Go
18h	DST_BEG_ADDR_SHADOW	Destination Begin Address Shadow Register	EALLOW	Go
1Ah	DST_ADDR_SHADOW	Destination Address Shadow Register	EALLOW	Go
1Ch	DST_BEG_ADDR_ACTIVE	Destination Begin Address Active Register	EALLOW	Go
1Eh	DST_ADDR_ACTIVE	Destination Address Active Register	EALLOW	Go

Complex bit access types are encoded to fit into small table cells. Table 5-11 shows the codes that are used for access types in this section.

Table 5-11. DMA_CH_REGS Access Type Codes

Access Type	Code	Description
Read Type		
R	R	Read
R-0	R -0	Read Returns 0s
Write Type		
W	W	Write
W1S	W 1S	Write 1 to set
Reset or Default Value		
-n		Value after reset or the default value

5.9.3.1 MODE Register (Offset = 0h) [Reset = 0000h]

MODE is shown in [Figure 5-13](#) and described in [Table 5-12](#).

Return to the [Summary Table](#).

Mode Register

Figure 5-13. MODE Register

15	14	13	12	11	10	9	8
CHINTE	DATASIZE	RESERVED	RESERVED	CONTINUOUS	ONESHOT	CHINTMODE	PERINTE
R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h	R/W-0h
7	6	5	4	3	2	1	0
OVRINTE	RESERVED		PERINTSEL				
R/W-0h	R-0h		R/W-0h				

Table 5-12. MODE Register Field Descriptions

Bit	Field	Type	Reset	Description
15	CHINTE	R/W	0h	Channel Interrupt Enable Bit This bit enables the DMA channel's CPU interrupt. Reset type: SYSRSn 0h (R/W) = Interrupt disabled 1h (R/W) = Interrupt enabled
14	DATASIZE	R/W	0h	Data Size Mode Bit This bit determines whether the DMA channel transfers 16 bits or 32 bits of data per read/write operation. Regardless of this setting, all data lengths and offsets in other DMA registers refer to 16-bit words. The pointer step increments must be configured to accommodate 32-bit words. Reset type: SYSRSn 0h (R/W) = 16-bit data transfer size 1h (R/W) = 32-bit data transfer size
13	RESERVED	R/W	0h	Reserved
12	RESERVED	R/W	0h	Reserved
11	CONTINUOUS	R/W	0h	Continuous Mode Bit If this bit is set to 1, then the channel re-initializes when TRANSFER_COUNT is zero and waits for the next event trigger. Otherwise, the DMA stops and clears the RUNSTS bit. Reset type: SYSRSn
10	ONESHOT	R/W	0h	One Shot Mode If this bit is set to 1, each peripheral event trigger causes the channel to perform an entire transfer. Otherwise, the channel only performs one burst per trigger. Reset type: SYSRSn
9	CHINTMODE	R/W	0h	Channel Interrupt Generation Mode This bit specifies when the DMA channel generates a CPU interrupt for a transfer. Reset type: SYSRSn 0h (R/W) = Generate interrupt at beginning of new transfer 1h (R/W) = Generate interrupt at end of transfer.
8	PERINTE	R/W	0h	Peripheral Event Trigger Enable This bit enables peripheral event triggers on the DMA channel. Reset type: SYSRSn 0h (R/W) = Peripheral event trigger disabled. Neither the selected peripheral nor software can start a DMA burst. 1h (R/W) = Peripheral event trigger enabled.

Table 5-12. MODE Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
7	OVRINTE	R/W	0h	Overflow Interrupt Enable The bit determines whether the DMA module generates a CPU interrupt when it detects an overflow event. Reset type: SYSRSn 0h (R/W) = Overflow interrupt disabled 1h (R/W) = Overflow interrupt enabled
6-5	RESERVED	R	0h	Reserved
4-0	PERINTSEL	R/W	0h	Peripheral Event Trigger Source Select These are legacy bits and should be set to the channel number. The actual source selection is done via the DMACHSRCSELn registers, which are part of the DMA_CLA_SRC_SEL_REGS group. Reset type: SYSRSn

5.9.3.2 CONTROL Register (Offset = 1h) [Reset = 0000h]

CONTROL is shown in [Figure 5-14](#) and described in [Table 5-13](#).

Return to the [Summary Table](#).

Control Register

Figure 5-14. CONTROL Register

15	14	13	12	11	10	9	8
RESERVED	OVRFLG	RUNSTS	BURSTSTS	TRANSFERSTS	RESERVED	RESERVED	PERINTFLG
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
7	6	5	4	3	2	1	0
ERRCLR	RESERVED	RESERVED	PERINTCLR	PERINTFRC	SOFTRESET	HALT	RUN
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h

Table 5-13. CONTROL Register Field Descriptions

Bit	Field	Type	Reset	Description
15	RESERVED	R	0h	Reserved
14	OVRFLG	R	0h	Overflow Flag This bit indicates that a peripheral event trigger was received while PERINTFLG was already set. It can be cleared by writing to the ERRCLR bit. Reset type: SYSRSn 0h (R/W) = No overflow detected 1h (R/W) = Overflow detected
13	RUNSTS	R	0h	Run Status Flag This bit indicates that the DMA channel is ready to respond to peripheral event triggers. This bit is set when a 1 is written to the RUN bit. It is cleared when a transfer completes (TRANSFER_COUNT = 0) and continuous mode is disabled, or when the HARDRESET, SOFTRESET, or HALT bit is set. Reset type: SYSRSn 0h (R/W) = The channel is disabled 1h (R/W) = The channel is enabled
12	BURSTSTS	R	0h	Burst Status Flag This bit is set when a DMA burst begins. The BURST_COUNT is set to the BURST_SIZE. This bit is cleared when BURST_COUNT reaches zero, or when the HARDRESET or SOFTRESET bit is set. Reset type: SYSRSn 0h (R/W) = No burst activity 1h (R/W) = The DMA is currently servicing or suspending a burst transfer from this channel
11	TRANSFERSTS	R	0h	Transfer Status Flag This bit is set when a DMA transfer begins. The address registers are copied to the shadow set and the TRANSFER_COUNT is set to the TRANSFER_SIZE. This bit is cleared when TRANSFER_COUNT reaches zero, or when the HARDRESET or SOFTRESET bit is set. Reset type: SYSRSn 0h (R/W) = No transfer activity 1h (R/W) = The channel is currently in the middle of a transfer regardless of whether a burst of data is actively being transferred or not
10	RESERVED	R	0h	Reserved
9	RESERVED	R	0h	Reserved

Table 5-13. CONTROL Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
8	PERINTFLG	R	0h	Peripheral Event Trigger Flag This bit indicates whether a peripheral event trigger has arrived. This bit is automatically cleared when the first burst transfer begins. Reset type: SYSRSn 0h (R/W) = Waiting for event trigger 1h (R/W) = Event trigger pending
7	ERRCLR	R-0/W1S	0h	Clear Error Writing a 1 to this bit will clear the OVRFLG bit. This is normally done when initializing the DMA module or if an overflow condition is detected. If an overflow event occurs at the same time this bit is set, the overrun has priority and the OVRFLG bit is set. Reset type: SYSRSn
6	RESERVED	R-0/W1S	0h	Reserved
5	RESERVED	R-0/W1S	0h	Reserved
4	PERINTCLR	R-0/W1S	0h	Clear Peripheral Event Trigger Writing a 1 to this bit clears PERINTFLG, which cancels a pending event trigger. This is normally done when initializing the DMA module. If an event trigger arrives at the same time this bit is set, the trigger has priority and PERINTFLG is set. Reset type: SYSRSn
3	PERINTFRC	R-0/W1S	0h	Force Peripheral Event Trigger If the PERINTE bit of the MODE register is set, writing a 1 to this bit sets PERINTFLG, which triggers a DMA burst. This bit can be used to start a DMA transfer in software. Reset type: SYSRSn
2	SOFTRESET	R-0/W1S	0h	Channel Soft Reset Writing a 1 to this bit places the channel into its default state after the current read/write access has completed: RUNSTS = 0 TRANSFERSTS = 0 BURSTSTS = 0 BURST_COUNT = 0 TRANSFER_COUNT = 0 SRC_WRAP_COUNT = 0 DST_WRAP_COUNT = 0 When writing to this bit, there is a one cycle delay before it takes effect. Hence, a one-cycle delay (such as a NOP instruction) is required in software before attempting to access any other DMA register. Reset type: SYSRSn
1	HALT	R-0/W1S	0h	Halt Channel Writing a 1 to this bit halts the DMA channel in its current state after any ongoing read/write access has completed. Reset type: SYSRSn
0	RUN	R-0/W1S	0h	Run Channel Writing a 1 to this bit enables the DMA channel and sets the RUNSTS bit to 1. This bit is also used to resume after a channel halt. The RUN bit is typically used to start the DMA channel after configuration. The channel will then wait for the first peripheral event trigger (PERINTFLG == 1) to start a burst. Reset type: SYSRSn

5.9.3.3 BURST_SIZE Register (Offset = 2h) [Reset = 0000h]

BURST_SIZE is shown in [Figure 5-15](#) and described in [Table 5-14](#).

Return to the [Summary Table](#).

Burst Size Register

Figure 5-15. BURST_SIZE Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				BURSTSIZE			
R-0h				R/W-0h			

Table 5-14. BURST_SIZE Register Field Descriptions

Bit	Field	Type	Reset	Description
15-5	RESERVED	R	0h	Reserved
4-0	BURSTSIZE	R/W	0h	These bits specify the burst size in 16-bit words. The actual size is equal to BURSTSIZE + 1. Reset type: SYSRSn

5.9.3.4 BURST_COUNT Register (Offset = 3h) [Reset = 0000h]

BURST_COUNT is shown in [Figure 5-16](#) and described in [Table 5-15](#).

Return to the [Summary Table](#).

Burst Count Register

Figure 5-16. BURST_COUNT Register

15	14	13	12	11	10	9	8
RESERVED							
R-0h							
7	6	5	4	3	2	1	0
RESERVED				BURSTCOUNT			
R-0h				R-0h			

Table 5-15. BURST_COUNT Register Field Descriptions

Bit	Field	Type	Reset	Description
15-5	RESERVED	R	0h	Reserved
4-0	BURSTCOUNT	R	0h	These bits indicate the number of words left in the current burst. Reset type: SYSRSn 0h (R/W) = 0 word left in a burst 1h (R/W) = 1 word left in a burst 2h (R/W) = 2 word left in a burst 3h (R/W) = 3 word left in a burst 4h (R/W) = 4 word left in a burst 5h (R/W) = 5 word left in a burst 6h (R/W) = 6 word left in a burst 7h (R/W) = 7 word left in a burst 8h (R/W) = 8 word left in a burst 9h (R/W) = 9 word left in a burst Ah (R/W) = 10 word left in a burst Bh (R/W) = 11 word left in a burst Ch (R/W) = 12 word left in a burst Dh (R/W) = 13 word left in a burst Eh (R/W) = 14 word left in a burst Fh (R/W) = 15 word left in a burst 10h (R/W) = 16 word left in a burst 11h (R/W) = 17 word left in a burst 12h (R/W) = 18 word left in a burst 13h (R/W) = 19 word left in a burst 14h (R/W) = 20 word left in a burst 15h (R/W) = 21 word left in a burst 16h (R/W) = 22 word left in a burst 17h (R/W) = 23 word left in a burst 18h (R/W) = 24 word left in a burst 19h (R/W) = 25 word left in a burst 1Ah (R/W) = 26 word left in a burst 1Bh (R/W) = 27 word left in a burst 1Ch (R/W) = 28 word left in a burst 1Dh (R/W) = 29 word left in a burst 1Eh (R/W) = 30 word left in a burst 1Fh (R/W) = 31 word left in a burst

5.9.3.5 SRC_BURST_STEP Register (Offset = 4h) [Reset = 0000h]

SRC_BURST_STEP is shown in [Figure 5-17](#) and described in [Table 5-16](#).

Return to the [Summary Table](#).

Source Burst Step Register

Figure 5-17. SRC_BURST_STEP Register

15	14	13	12	11	10	9	8
SRCBURSTSTEP							
R/W-0h							
7	6	5	4	3	2	1	0
SRCBURSTSTEP							
R/W-0h							

Table 5-16. SRC_BURST_STEP Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	SRCBURSTSTEP	R/W	0h	These bits specify the change in the source address after each word in a burst. The size must be a 16-bit two's complement value between -4096 and 4095 (inclusive). This value is added to the source address after each read/write operation in a burst. Reset type: SYSRSn 0h (R/W) = No address change 1h (R/W) = Add 1 to the address 2h (R/W) = Add 2 to the address FFEh (R/W) = Add 4094 to the address FFFh (R/W) = Add 4095 to the address F000h (R/W) = Subtract 4096 from the address F001h (R/W) = Subtract 4095 from the address FFFEh (R/W) = Subtract 2 from the address FFFFh (R/W) = Subtract 1 from the address

5.9.3.6 DST_BURST_STEP Register (Offset = 5h) [Reset = 0000h]

DST_BURST_STEP is shown in [Figure 5-18](#) and described in [Table 5-17](#).

Return to the [Summary Table](#).

Destination Burst Step Register

Figure 5-18. DST_BURST_STEP Register

15	14	13	12	11	10	9	8
DSTBURSTSTEP							
R/W-0h							
7	6	5	4	3	2	1	0
DSTBURSTSTEP							
R/W-0h							

Table 5-17. DST_BURST_STEP Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	DSTBURSTSTEP	R/W	0h	These bits specify the change in the destination address after each word in a burst. The size must be a 16-bit two's complement value between -4096 and 4095 (inclusive). This value is added to the destination address after each read/write operation in a burst. Reset type: SYSRSn 0h (R/W) = No address change 1h (R/W) = Add 1 to the address 2h (R/W) = Add 2 to the address FFEh (R/W) = Add 4094 to the address FFFh (R/W) = Add 4095 to the address F000h (R/W) = Subtract 4096 from the address F001h (R/W) = Subtract 4095 from the address FFFEh (R/W) = Subtract 2 from the address FFFFh (R/W) = Subtract 1 from the address

5.9.3.7 TRANSFER_SIZE Register (Offset = 6h) [Reset = 0000h]

TRANSFER_SIZE is shown in [Figure 5-19](#) and described in [Table 5-18](#).

Return to the [Summary Table](#).

Transfer Size Register

Figure 5-19. TRANSFER_SIZE Register

15	14	13	12	11	10	9	8
TRANSFERSIZE							
R/W-0h							
7	6	5	4	3	2	1	0
TRANSFERSIZE							
R/W-0h							

Table 5-18. TRANSFER_SIZE Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	TRANSFERSIZE	R/W	0h	These bits specify the transfer size in bursts. The actual size is equal to TRANSFERSIZE + 1. Reset type: SYSRSn

5.9.3.8 TRANSFER_COUNT Register (Offset = 7h) [Reset = 0000h]

TRANSFER_COUNT is shown in [Figure 5-20](#) and described in [Table 5-19](#).

Return to the [Summary Table](#).

Transfer Count Register

Figure 5-20. TRANSFER_COUNT Register

15	14	13	12	11	10	9	8
TRANSFERCOUNT							
R-0h							
7	6	5	4	3	2	1	0
TRANSFERCOUNT							
R-0h							

Table 5-19. TRANSFER_COUNT Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	TRANSFERCOUNT	R	0h	These bits indicate the number of bursts left in the current transfer. Reset type: SYSRSn

5.9.3.9 SRC_TRANSFER_STEP Register (Offset = 8h) [Reset = 0000h]

SRC_TRANSFER_STEP is shown in [Figure 5-21](#) and described in [Table 5-20](#).

Return to the [Summary Table](#).

Source Transfer Step Register

Figure 5-21. SRC_TRANSFER_STEP Register

15	14	13	12	11	10	9	8
SRCTransferStep							
R/W-0h							
7	6	5	4	3	2	1	0
SRCTransferStep							
R/W-0h							

Table 5-20. SRC_TRANSFER_STEP Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	SRCTransferStep	R/W	0h	These bits specify the change in the source address after a burst completes. The size must be a 16-bit two's complement value between -4096 and 4095 (inclusive). This value is added to the source address after each burst completes. Reset type: SYSRSn 0h (R/W) = No address change 1h (R/W) = Add 1 to the address 2h (R/W) = Add 2 to the address FFEh (R/W) = Add 4094 to the address FFFh (R/W) = Add 4095 to the address F000h (R/W) = Subtract 4096 from the address F001h (R/W) = Subtract 4095 from the address FFFEh (R/W) = Subtract 2 from the address FFFFh (R/W) = Subtract 1 from the address

5.9.3.10 DST_TRANSFER_STEP Register (Offset = 9h) [Reset = 0000h]

DST_TRANSFER_STEP is shown in [Figure 5-22](#) and described in [Table 5-21](#).

Return to the [Summary Table](#).

Destination Transfer Step Register

Figure 5-22. DST_TRANSFER_STEP Register

15	14	13	12	11	10	9	8
DSTTRANSFERSTEP							
R/W-0h							
7	6	5	4	3	2	1	0
DSTTRANSFERSTEP							
R/W-0h							

Table 5-21. DST_TRANSFER_STEP Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	DSTTRANSFERSTEP	R/W	0h	These bits specify the change in the destination address after a burst completes. The size must be a 16-bit two's complement value between -4096 and 4095 (inclusive). This value is added to the destination address after each burst completes. Reset type: SYSRSn 0h (R/W) = No address change 1h (R/W) = Add 1 to the address 2h (R/W) = Add 2 to the address FFEh (R/W) = Add 4094 to the address FFFh (R/W) = Add 4095 to the address F000h (R/W) = Subtract 4096 from the address F001h (R/W) = Subtract 4095 from the address FFFEh (R/W) = Subtract 2 from the address FFFFh (R/W) = Subtract 1 from the address

5.9.3.11 SRC_WRAP_SIZE Register (Offset = Ah) [Reset = FFFFh]

SRC_WRAP_SIZE is shown in [Figure 5-23](#) and described in [Table 5-22](#).

Return to the [Summary Table](#).

Source Wrap Size Register

Figure 5-23. SRC_WRAP_SIZE Register

15	14	13	12	11	10	9	8
WRAPSIZE							
R/W-FFFFh							
7	6	5	4	3	2	1	0
WRAPSIZE							
R/W-FFFFh							

Table 5-22. SRC_WRAP_SIZE Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	WRAPSIZE	R/W	FFFFh	These bits specify the number of bursts to transfer before the source address wraps around to the beginning address. The actual number is equal to WRAPSIZE + 1. To disable the wrapping function, set WRAPSIZE to a value larger than TRANSFERSIZE. Reset type: SYSRSn

5.9.3.12 SRC_WRAP_COUNT Register (Offset = Bh) [Reset = 0000h]

SRC_WRAP_COUNT is shown in [Figure 5-24](#) and described in [Table 5-23](#).

Return to the [Summary Table](#).

Source Wrap Count Register

Figure 5-24. SRC_WRAP_COUNT Register

15	14	13	12	11	10	9	8
WRAPSIZE							
R-0h							
7	6	5	4	3	2	1	0
WRAPSIZE							
R-0h							

Table 5-23. SRC_WRAP_COUNT Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	WRAPSIZE	R	0h	These bits indicate the number of bursts left before wrapping the source address. Reset type: SYSRSn

5.9.3.13 SRC_WRAP_STEP Register (Offset = Ch) [Reset = 0000h]

SRC_WRAP_STEP is shown in [Figure 5-25](#) and described in [Table 5-24](#).

Return to the [Summary Table](#).

Source Wrap Step Register

Figure 5-25. SRC_WRAP_STEP Register

15	14	13	12	11	10	9	8
WRAPSTEP							
R/W-0h							
7	6	5	4	3	2	1	0
WRAPSTEP							
R/W-0h							

Table 5-24. SRC_WRAP_STEP Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	WRAPSTEP	R/W	0h	These bits specify the change in the source beginning address when the wrap counter reaches zero. The size must be a 16-bit two's complement value between -4096 and 4095 (inclusive). This value is added to the source address when wrapping occurs. Reset type: SYSRSn 0h (R/W) = No address change 1h (R/W) = Add 1 to the address 2h (R/W) = Add 2 to the address FFEh (R/W) = Add 4094 to the address FFFh (R/W) = Add 4095 to the address F000h (R/W) = Subtract 4096 from the address F001h (R/W) = Subtract 4095 from the address FFFEh (R/W) = Subtract 2 from the address FFFFh (R/W) = Subtract 1 from the address

5.9.3.14 DST_WRAP_SIZE Register (Offset = Dh) [Reset = FFFFh]

DST_WRAP_SIZE is shown in [Figure 5-26](#) and described in [Table 5-25](#).

Return to the [Summary Table](#).

Destination Wrap Size Register

Figure 5-26. DST_WRAP_SIZE Register

15	14	13	12	11	10	9	8
WRAPSIZE							
R/W-FFFFh							
7	6	5	4	3	2	1	0
WRAPSIZE							
R/W-FFFFh							

Table 5-25. DST_WRAP_SIZE Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	WRAPSIZE	R/W	FFFFh	These bits specify the number of bursts to transfer before the destination address wraps around to the beginning address. The actual number is equal to WRAPSIZE + 1. To disable the wrapping function, set WRAPSIZE to a value larger than TRANSFERSIZE. Reset type: SYSRSn

5.9.3.15 DST_WRAP_COUNT Register (Offset = Eh) [Reset = 0000h]

DST_WRAP_COUNT is shown in [Figure 5-27](#) and described in [Table 5-26](#).

Return to the [Summary Table](#).

Destination Wrap Count Register

Figure 5-27. DST_WRAP_COUNT Register

15	14	13	12	11	10	9	8
WRAPSIZE							
R-0h							
7	6	5	4	3	2	1	0
WRAPSIZE							
R-0h							

Table 5-26. DST_WRAP_COUNT Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	WRAPSIZE	R	0h	These bits indicate the number of bursts left before wrapping the destination address. Reset type: SYSRSn

5.9.3.16 DST_WRAP_STEP Register (Offset = Fh) [Reset = 0000h]

DST_WRAP_STEP is shown in [Figure 5-28](#) and described in [Table 5-27](#).

Return to the [Summary Table](#).

Destination Wrap Step Register

Figure 5-28. DST_WRAP_STEP Register

15	14	13	12	11	10	9	8
WRAPSTEP							
R/W-0h							
7	6	5	4	3	2	1	0
WRAPSTEP							
R/W-0h							

Table 5-27. DST_WRAP_STEP Register Field Descriptions

Bit	Field	Type	Reset	Description
15-0	WRAPSTEP	R/W	0h	These bits specify the change in the destination beginning address when the wrap counter reaches zero. The size must be a 16-bit two's complement value between -4096 and 4095 (inclusive). This value is added to the destination address when wrapping occurs. Reset type: SYSRSn 0h (R/W) = No address change 1h (R/W) = Add 1 to the address 2h (R/W) = Add 2 to the address FFEh (R/W) = Add 4094 to the address FFFh (R/W) = Add 4095 to the address F000h (R/W) = Subtract 4096 from the address F001h (R/W) = Subtract 4095 from the address FFFEh (R/W) = Subtract 2 from the address FFFFh (R/W) = Subtract 1 from the address

5.9.3.17 SRC_BEG_ADDR_SHADOW Register (Offset = 10h) [Reset = 00000000h]

SRC_BEG_ADDR_SHADOW is shown in [Figure 5-29](#) and described in [Table 5-28](#).

Return to the [Summary Table](#).

Source Begin Address Shadow Register

Figure 5-29. SRC_BEG_ADDR_SHADOW Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BEGADDR																															
R/W-0h																															

Table 5-28. SRC_BEG_ADDR_SHADOW Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	BEGADDR	R/W	0h	Shadow Source Beginning Address At the start of a transfer, the value in this register is loaded into the SRC_BEG_ADDR_ACTIVE register and used as the beginning value for the source address. This register can be safely updated during a transfer. Reset type: SYSRSn

5.9.3.18 SRC_ADDR_SHADOW Register (Offset = 12h) [Reset = 00000000h]

SRC_ADDR_SHADOW is shown in [Figure 5-30](#) and described in [Table 5-29](#).

Return to the [Summary Table](#).

Source Address Shadow Register

Figure 5-30. SRC_ADDR_SHADOW Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR																															
R/W-0h																															

Table 5-29. SRC_ADDR_SHADOW Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	ADDR	R/W	0h	Shadow Source Address At the start of a transfer, the value in this register is loaded into the SRC_ADDR_ACTIVE register and used as the value of the source address. This register can be safely updated during a transfer. Reset type: SYSRSn

5.9.3.19 SRC_BEG_ADDR_ACTIVE Register (Offset = 14h) [Reset = 00000000h]

SRC_BEG_ADDR_ACTIVE is shown in [Figure 5-31](#) and described in [Table 5-30](#).

Return to the [Summary Table](#).

Source Begin Address Active Register

Figure 5-31. SRC_BEG_ADDR_ACTIVE Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BEGADDR																															
R-0h																															

Table 5-30. SRC_BEG_ADDR_ACTIVE Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	BEGADDR	R	0h	Active Source Beginning Address If a transfer is ongoing, this register holds the current beginning value for the source address. This address may be updated after wrapping. When a transfer starts, this register is loaded with the shadow address from the SRC_BEG_ADDR_SHADOW register. Reset type: SYSRSn

5.9.3.20 SRC_ADDR_ACTIVE Register (Offset = 16h) [Reset = 00000000h]

SRC_ADDR_ACTIVE is shown in [Figure 5-32](#) and described in [Table 5-31](#).

Return to the [Summary Table](#).

Source Address Active Register

Figure 5-32. SRC_ADDR_ACTIVE Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR																															
R-0h																															

Table 5-31. SRC_ADDR_ACTIVE Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	ADDR	R	0h	Active Source Address If a transfer is ongoing, this register holds the current value of the source address. This address may change after a write, a burst, or wrapping. Reset type: SYSRSn

5.9.3.21 DST_BEG_ADDR_SHADOW Register (Offset = 18h) [Reset = 00000000h]

DST_BEG_ADDR_SHADOW is shown in [Figure 5-33](#) and described in [Table 5-32](#).

Return to the [Summary Table](#).

Destination Begin Address Shadow Register

Figure 5-33. DST_BEG_ADDR_SHADOW Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BEGADDR																															
R/W-0h																															

Table 5-32. DST_BEG_ADDR_SHADOW Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	BEGADDR	R/W	0h	Shadow Destination Beginning Address At the start of a transfer, the value in this register is loaded into the DST_BEG_ADDR_ACTIVE register and used as the beginning value for the destination address. This register can be safely updated during a transfer. Reset type: SYSRSn

5.9.3.22 DST_ADDR_SHADOW Register (Offset = 1Ah) [Reset = 00000000h]

DST_ADDR_SHADOW is shown in [Figure 5-34](#) and described in [Table 5-33](#).

Return to the [Summary Table](#).

Destination Address Shadow Register

Figure 5-34. DST_ADDR_SHADOW Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR																															
R/W-0h																															

Table 5-33. DST_ADDR_SHADOW Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	ADDR	R/W	0h	Shadow Destination Address At the start of a transfer, the value in this register is loaded into the DST_ADDR_ACTIVE register and used as the value of the destination address. This register can be safely updated during a transfer. Reset type: SYSRSn

5.9.3.23 DST_BEG_ADDR_ACTIVE Register (Offset = 1Ch) [Reset = 00000000h]

DST_BEG_ADDR_ACTIVE is shown in [Figure 5-35](#) and described in [Table 5-34](#).

Return to the [Summary Table](#).

Destination Begin Address Active Register

Figure 5-35. DST_BEG_ADDR_ACTIVE Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BEGADDR																															
R-0h																															

Table 5-34. DST_BEG_ADDR_ACTIVE Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	BEGADDR	R	0h	Active Destination Beginning Address If a transfer is ongoing, this register holds the current destination value for the source address. This address may be updated after wrapping. When a transfer starts, this register is loaded with the shadow address from the DST_BEG_ADDR_SHADOW register. Reset type: SYSRSn

5.9.3.24 DST_ADDR_ACTIVE Register (Offset = 1Eh) [Reset = 00000000h]

DST_ADDR_ACTIVE is shown in [Figure 5-36](#) and described in [Table 5-35](#).

Return to the [Summary Table](#).

Destination Address Active Register

Figure 5-36. DST_ADDR_ACTIVE Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR																															
R-0h																															

Table 5-35. DST_ADDR_ACTIVE Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	ADDR	R	0h	Active Destination Address If a transfer is ongoing, this register holds the current value of the destination address. This address may change after a write, a burst, or wrapping. Reset type: SYSRSn

5.9.4 DMA Registers to Driverlib Functions

Table 5-36. DMA Registers to Driverlib Functions

File	Driverlib Function
CTRL	
dma.h	DMA_initController
DEBUGCTRL	
dma.h	DMA_setEmulationMode
PRIORITYCTRL1	
dma.h	DMA_setPriorityMode
PRIORITYSTAT	
-	
MODE	
dma.c	DMA_configMode
dma.h	DMA_enableTrigger
dma.h	DMA_disableTrigger
dma.h	DMA_enableInterrupt
dma.h	DMA_disableInterrupt
dma.h	DMA_enableOverrunInterrupt
dma.h	DMA_disableOverrunInterrupt
dma.h	DMA_setInterruptMode
CONTROL	
dma.h	DMA_triggerSoftReset
dma.h	DMA_forceTrigger
dma.h	DMA_clearTriggerFlag
dma.h	DMA_getTransferStatusFlag
dma.h	DMA_getBurstStatusFlag
dma.h	DMA_getRunStatusFlag
dma.h	DMA_getOverflowFlag
dma.h	DMA_getTriggerFlagStatus
dma.h	DMA_startChannel

Table 5-36. DMA Registers to Driverlib Functions (continued)

File	Driverlib Function
dma.h	DMA_stopChannel
dma.h	DMA_clearErrorFlag
BURST_SIZE	
dma.c	DMA_configBurst
BURST_COUNT	
-	
SRC_BURST_STEP	
dma.c	DMA_configBurst
DST_BURST_STEP	
dma.c	DMA_configBurst
TRANSFER_SIZE	
dma.c	DMA_configTransfer
TRANSFER_COUNT	
-	
SRC_TRANSFER_STEP	
dma.c	DMA_configTransfer
DST_TRANSFER_STEP	
dma.c	DMA_configTransfer
SRC_WRAP_SIZE	
dma.c	DMA_configWrap
SRC_WRAP_COUNT	
-	
SRC_WRAP_STEP	
dma.c	DMA_configWrap
DST_WRAP_SIZE	
dma.c	DMA_configWrap
DST_WRAP_COUNT	
-	
DST_WRAP_STEP	
dma.c	DMA_configWrap
SRC_BEG_ADDR_SHADOW	
dma.c	DMA_configAddresses
dma.h	DMA_configSourceAddress
SRC_ADDR_SHADOW	
dma.c	DMA_configAddresses
dma.h	DMA_configSourceAddress
SRC_BEG_ADDR_ACTIVE	
-	
SRC_ADDR_ACTIVE	
-	
DST_BEG_ADDR_SHADOW	
dma.c	DMA_configAddresses
dma.h	DMA_configDestAddress
DST_ADDR_SHADOW	
dma.c	DMA_configAddresses

Table 5-36. DMA Registers to Driverlib Functions (continued)

File	Driverlib Function
dma.h	DMA_configDestAddress
DST_BEG_ADDR_ACTIVE	
-	
DST_ADDR_ACTIVE	
-	